Second MU Calculation
Oral Exam Master Packet

Color-coded study guide with derivations, figures, clinical traps, and product notes

Purpose

This packet is written to be the kind of document you can *study from directly* and also *speak from in an oral exam*. It is intentionally organized around the questions examiners tend to ask:

- What is the full MU equation and what does every symbol mean?
- Why do we do a second check, and what can it actually catch?
- When do hand checks or secondary checks fail?
- How do heterogeneity, wedges, blocked fields, skin collimation, and extended SSD change the answer?
- What do TG-114 and TG-219 really expect you to do?

How to use this packet

Read the blue and green boxes first. Memorize the derivations in the *Derivations* section. Then read the *Rapid-fire oral exam answers* section out loud until the language becomes natural.

Prepared as a self-contained LaTeX study document

No TikZ required; figures are shown as lightweight boxed diagrams

Contents

1 Why a second MU check exists

High-yield point

A second MU check is *not* a replacement for commissioning, TPS QA, or a full plan review. It is an independent verification that the machine settings computed by the primary system are plausible and will deliver the intended dose at a representative point or, in more advanced systems, across a larger patient volume.[2, 3]

Oral-exam answer

Why do we do a second check?

Because history shows that independent verification catches clinically meaningful problems. TG-114 explicitly discusses treatment incidents in which loss or absence of an independent calculation contributed to patient overdose; it also stresses that modern TPS complexity *increases*, not decreases, the need for a well-designed independent verification process.[2]

For IMRT/VMAT, TG-219 extends that logic: the secondary calculation remains important, but it must be understood as one component of a *comprehensive* QA program and *not* a substitute for measurement-based PSQA.[3]

Common trap

The oral-exam trap: "The second check proves the plan is correct."

That is wrong. A second check can show that the primary answer is plausible at the chosen point or in the chosen comparison metric. It does *not* prove the entire 3D dose distribution is correct, and a poorly designed second check can reproduce the same wrong assumptions as the TPS.[2]

Clinical workflow

What questions should your head go through when you compare TPS vs second check?

1. Is the comparison point well chosen?
2. Are the field size definitions for S_c, S_p, and depth terms consistent?
3. Are all modifiers included: wedge, tray, bolus, compensator, cutout, skin collimation?
4. Is heterogeneity handled in both systems?
5. Is the disagreement explainable by known algorithm limitations?
6. If not, do I have a real physics problem, a modeling problem, or a data-transfer problem?

1.1 What a second check can catch

TG-114 lists a broad set of potential error types that second-check workflows may help expose, including wrong or omitted trays, wrong wedge or wedge direction, wrong energy, wrong field size, wrong beam weights, wrong depth, wrong SSD, poor point placement near field edges, wrong CT dataset, incorrect density mapping, data transfer errors, using untested calculation modules, and software bugs.[2]

Key formula

Keep this sentence ready for the oral exam:

A second MU check is best at catching gross plausibility problems, missing factors, wrong geometry, wrong depth/SSD, wrong field-size interpretation, wrong accessories, some beam-model problems, and some data-transfer or software issues. It is much weaker in high-gradient, interface, small-field, and highly modulated situations unless the secondary method is itself 2D/3D and robustly commissioned.

1.2 What if you do not have a second MU check

TG-114 recognizes that either an *independent calculation* or an *in-phantom dose measurement* can serve as a valid MU verification method.[2] If neither is available before treatment, the department should have a written emergency policy. TG-114 recommends completing the verification before the first fraction whenever possible; if not, it should be completed as soon as possible and no later than the earlier of the third fraction or 10% of the prescribed dose. For after-hours emergency treatment, TG-114 suggests placing a departmental limit on the untreated dose that can be delivered before full physics review, and quotes 3–4 Gy as a reasonable policy value.[2]

2 Core photon and electron MU equations

2.1 Photon MU equations

Key formula

Photon MU equation in TPR formalism (isocentric / SAD style)

$$\mathrm{MU} = \frac{D}{\dot{D}_0\ S_c(r_c)\ S_p(r_d)\ TPR(d, r_d)\ WF(d, r_d, x)\ TF\ OAR(d, x)\ \left(\dfrac{SSD_0 + d_0}{SPD}\right)^2} \tag{1}$$

If the point of calculation is at isocenter, then $SPD = SAD$, and the same equation becomes

$$\mathrm{MU} = \frac{D}{\dot{D}_0\ S_c(r_c)\ S_p(r_d)\ TPR(d, r_d)\ WF(d, r_d)\ TF\ \left(\dfrac{SSD_0 + d_0}{SAD}\right)^2}. \tag{2}$$

Key formula

Photon MU equation in PDD formalism (SSD style)

$$\mathrm{MU} = \frac{D \cdot 100}{\dot{D}_0\ S_c(r_c)\ S_p(r_{d_0})\ PDD_N(d, r, SSD)\ WF(d, r_d, x)\ TF\ OAR(d, x)\ \left(\dfrac{SSD_0 + d_0}{SSD + d_0}\right)^2}. \tag{3}$$

Definition

What $\dot{D}_0$ means

$\dot{D}_0$ is the dose per MU under the beam's normalization conditions. In TG-71 language, this is the dose-per-MU reference quantity for the beam under your department's chosen calibration geometry.[1]

2.2 Electron MU equations

Key formula

Electron MU equation at nominal SSD

$$\text{MU} = \frac{D \cdot 100}{\dot{D}_0 \; PDD(d, r_a, SSD_0) \; S_e(r_a, SSD_0)}. \tag{4}$$

Key formula

Electron MU equation at extended SSD using effective SSD

$$\text{MU} = \frac{D \cdot 100}{\dot{D}_0 \; PDD(d, r_a, SSD) \; S_e(r_a, SSD_0) \; \left(\frac{SSD_{\text{eff}}(r_a) + d_0}{SSD_{\text{eff}}(r_a) + d_0 + g}\right)^2}, \qquad g = SSD - SSD_0. \tag{5}$$

Key formula

Electron MU equation at extended SSD using air-gap factor

$$\text{MU} = \frac{D \cdot 100}{\dot{D}_0 \; PDD(d, r_a, SSD) \; S_e(r_a, SSD_0) \; \left(\frac{SSD_0 + d_0}{SSD_0 + d_0 + g}\right)^2 \; f_{\text{air}}(r_a, SSD)}. \tag{6}$$

2.3 Parameter dictionary

Symbol	Name	What to say in the oral exam
D	Prescribed dose to the point	The dose I want at the calculation point for that beam contribution.
$\dot{D}_0$	Dose per MU at normalization conditions	The beam output under reference conditions used to normalize all the other factors.
r_c	Collimator field size argument	The equivalent square defined by the variable aperture *closest to the source* that controls head scatter for the machine design.[1]
r_d	Field size at the calculation depth	The equivalent square of the field *incident on the patient*, projected to the calculation depth for TPR/S_p style quantities.[1]
r_{d_0}	Field size at normalization depth	Same concept as r_d, but projected to d_0 for SSD/PDD formalism.[1]

Symbol	Name	What to say in the oral exam
S_c	Collimator scatter factor	In-air output ratio; head scatter only. Measured in a mini-phantom.[1]
S_p	Phantom scatter factor	Scatter from the irradiated phantom volume only, defined for the field size at the point; $S_p = S_{c,p}/S_c$.[1]
$S_{c,p}$	Total scatter factor	In-water output ratio that combines head and phantom scatter.
PDD or PDD_N	Percent depth dose	SSD-based depth-dose quantity. Depends on energy, field size, and SSD.[1, 6]
TPR/TMR	Tissue-phantom / tissue-maximum ratio	SAD-style depth-dose ratio; removes direct SSD dependence from the setup geometry.[1, 6]
WF	Wedge factor	Ratio of dose with wedge to dose without wedge at the calculation point. In TG-71, all wedge-related beam hardening and scatter effects are absorbed into this single factor.[1]
TF	Tray factor	Ratio of dose with tray to dose without tray at the calculation point; nearly constant across field size/depth for many trays, but must be measured.[1]
OAR	Off-axis ratio	Used when the calculation point is off-axis. TG-71 prefers primary off-axis ratios over raw large-field profiles, especially near field edges.[1]
CF	Correction factor	In MU hand calculations this is most often the catch-all term for heterogeneity, missing tissue, or other geometry corrections.
S_e	Electron output factor	Electron output for the applicator/insert combination, typically at d_m, relative to the reference electron field.[1]
SSD_{eff}	Effective SSD	The virtual electron source distance that best fits extended-SSD output behavior for the field and energy.[1]
f_{air}	Electron air-gap factor	The multiplicative correction that accounts for the fact that extended-SSD electron output drops faster than pure inverse square because of side-scatter loss.[1]
BSF	Backscatter factor	Mostly historical in MV hand-calc conversation. In kilovoltage and classical TAR language, BSF is the low-energy special case of peak scatter factor. In modern MV photon MU checks, the more common formalism is S_c, S_p, and $S_{c,p}$.[6, 1]

2.4 A note on BSF

Definition

If you get asked: "What is BSF?"
A safe, high-level answer is:
Backscatter factor is a classical scatter quantity used most naturally with TAR-style or kilovoltage dosimetry. In modern megavoltage external-beam MU calculations, we usually work in the S_c*–*S_p*–*$S_{c,p}$ *formalism instead. If my department uses the word BSF on a worksheet, I would clarify whether they mean true backscatter factor, peak scatter factor, or a local shorthand at* $d_{\max}$.[6, 1]

2.5 Geometry picture

Conceptual Geometry Diagram

- Source-to-surface distance (SSD)
- Source-to-axis distance (SAD)
- Source-to-point distance (SPD)
- Calculation depth d and reference depth d_0

Figure 1: SSD, SAD, and source-to-point geometry for photon MU calculations.

3 Derivations you should be able to talk through

3.1 Inverse square law

Key formula

Basic physics
For an isotropic source or an effective point source, fluence falls as

$$\Phi \propto \frac{1}{r^2}. \tag{7}$$

Therefore, if every other factor is constant,

$$\frac{D_2}{D_1} \approx \left(\frac{r_1}{r_2}\right)^2. \tag{8}$$

Oral-exam answer

How do you explain "inverse square law" clinically?
At larger distance the beam cross-sectional area is larger, so the same number of photons is spread over a bigger area. Dose therefore falls like $1/r^2$, or more precisely, the primary fluence component does. In photon MU equations that geometry is explicit in the $\left((SSD_0 + d_0)/(\text{distance})\right)^2$ term.[1]

3.2 Extended SSD for photons

In SSD style photon calculations, the explicit inverse-square term is

$$ISL_{\text{photon, SSD}} = \left(\frac{SSD_0 + d_0}{SSD + d_0}\right)^2. \tag{9}$$

That term *only* handles the change in intensity from the change in reference-point distance. It does *not* automatically account for the fact that percent depth dose itself changes with SSD. That extra change is why Mayneord's factor was introduced.

3.3 Mayneord factor derivation

Key formula

Definition

Mayneord's F-factor is the classic approximation that converts a PDD value from one SSD to another *by accounting only for the inverse-square part of the change.*[5, 6]

Start from the definition

$$PDD(d, f) = 100 \times \frac{D(d, f)}{D(d_{\max}, f)}, \tag{10}$$

where f denotes SSD.

Now assume that the attenuation/scatter ratio is unchanged when moving from f_1 to f_2. Then depth dose at a point varies only with source-to-point distance:

$$D(d, f) \propto \frac{K_d}{(f + d)^2}, \qquad D(d_{\max}, f) \propto \frac{K_{\max}}{(f + d_{\max})^2}. \tag{11}$$

Therefore,

$$PDD(d, f) = 100 \times \frac{K_d}{K_{\max}} \left(\frac{f + d_{\max}}{f + d}\right)^2. \tag{12}$$

Take the ratio of the two PDDs:

$$\frac{PDD(d, f_2)}{PDD(d, f_1)} = \frac{100\frac{K_d}{K_{\max}}\left(\frac{f_2+d_{\max}}{f_2+d}\right)^2}{100\frac{K_d}{K_{\max}}\left(\frac{f_1+d_{\max}}{f_1+d}\right)^2} \tag{13}$$

$$= \left(\frac{f_2 + d_{\max}}{f_2 + d}\right)^2 \left(\frac{f_1 + d}{f_1 + d_{\max}}\right)^2. \tag{14}$$

So the Mayneord factor is

$$F_M = \left(\frac{f_2 + d_{\max}}{f_2 + d}\right)^2 \left(\frac{f_1 + d}{f_1 + d_{\max}}\right)^2, \tag{15}$$

and the conversion is

$$PDD(d, f_2) \approx PDD(d, f_1)\, F_M. \tag{16}$$

Common trap

What Mayneord misses

Mayneord handles *geometry only*. It ignores the fact that scatter conditions also change with SSD. For modest SSD changes it is often acceptable as a hand-check approximation; for very extended distances, such as TBI geometries, the error can be several percent, so direct measurement is preferred.[5]

3.4 PDD to TMR conversion derivation

Key formula

Statement of the result

If the corresponding field sizes are handled consistently, then

$$TMR(d, r_d) = \frac{PDD(d, r, SSD)}{100} \left(\frac{SSD + d}{SSD + d_{\max}} \right)^2 \frac{S_p(r_{d_{\max}})}{S_p(r_d)}. \tag{17}$$

Derivation. At depth d, a photon dose model in ratio form can be written schematically as

$$D(d) \propto \frac{S_p(r_d)\, TMR(d, r_d)}{(SSD + d)^2}. \tag{18}$$

At $d_{\max}$, since $TMR(d_{\max}, r_{d_{\max}}) = 1$,

$$D(d_{\max}) \propto \frac{S_p(r_{d_{\max}})}{(SSD + d_{\max})^2}. \tag{19}$$

Take the ratio:

$$\frac{PDD(d, r, SSD)}{100} = \frac{D(d)}{D(d_{\max})} \tag{20}$$

$$= \frac{S_p(r_d)\, TMR(d, r_d)/(SSD + d)^2}{S_p(r_{d_{\max}})/(SSD + d_{\max})^2} \tag{21}$$

$$= TMR(d, r_d)\, \frac{S_p(r_d)}{S_p(r_{d_{\max}})} \left(\frac{SSD + d_{\max}}{SSD + d} \right)^2. \tag{22}$$

Rearranging gives Eq. (17).

Oral-exam answer

How do you explain this derivation out loud?

PDD and TMR are both depth-dose ratios, but they normalize to *different things*. PDD normalizes to dose at $d_{\max}$ at fixed SSD, so it contains inverse-square effects automatically. TMR compares doses at the *same source-to-detector distance* in phantom, so the geometric $1/r^2$ effect has to be removed. Then you still need a scatter correction, because the field projected to d is not the same as the field projected to $d_{\max}$. That is what the $S_p(r_{d_{\max}})/S_p(r_d)$ ratio does.[1, 7]

Definition

Where does BSF show up in older derivations?

In older TAR/BSF derivations, the normalization at $d_{\max}$ is often handled using a peak-scatter or backscatter quantity instead of modern S_p notation. That is why some textbooks and quick references write the conversion with BSF or PSF rather than a phantom-scatter ratio.[7, 6]

3.5 Extended SSD for electrons

Electrons are *not* pure inverse-square beams at extended SSD. Side scatter is lost, and the loss is worse for smaller inserts and lower energies.[1] That is why TG-71 includes both SSD_{eff} and f_{air} methods.

Extended-SSD Electron Diagram

- Nominal surface at SSD_0
- Extended SSD surface separated by air gap g
- Effective SSD / air-gap correction
- Loss of lateral scatter at extended distance

Figure 2: Extended-SSD electron geometry. The dose drop is steeper than $1/r^2$ when lateral scatter equilibrium is degraded.

3.6 Sensitivity of MU to factor changes

Key formula

General sensitivity rule

If the MU equation is of the form

$$\text{MU} = \frac{D}{f_1 f_2 f_3 \cdots},$$

then the first-order fractional sensitivity is

$$\frac{\Delta\text{MU}}{\text{MU}} \approx \frac{\Delta D}{D} - \sum_i \frac{\Delta f_i}{f_i}. \tag{23}$$

Oral-exam answer

What does that mean clinically?

It means most multiplicative factors are almost perfectly one-for-one in the opposite direction:

- If S_c is 1% higher than you thought, MU is about 1% lower.
- If WF is 3% lower than you thought, MU is about 3% higher.
- If you omit a tray factor of 0.97, you are roughly making a 3% mistake.

Distance errors can be even more dangerous. If the relevant source-to-point distance is R, then from inverse square

$$\frac{\Delta\text{MU}}{\text{MU}} \approx 2\frac{\Delta R}{R}.$$

So a 1 cm error at $R = 90$ cm is about a 2.2% MU error before you even think about PDD/TMR changes.

MU Sensitivity Summary

- Most multiplicative factor errors translate nearly one-for-one into MU error
- A 1% error in S_c, S_p, PDD, TMR, WF, or TF gives about a 1% MU change in the opposite direction
- Distance errors can be larger because inverse-square effects scale approximately as $2\Delta R/R$

Figure 3: Product-form MU equations are almost one-for-one sensitive to multiplicative factor errors.

4 Field size logic, trends, and what changes with machine design

4.1 Which field size goes into which factor

Oral-exam answer

This is one of the most important oral-exam answers.

- S_c: use the field size defined by the aperture system *closest to the source* that controls head scatter.[1]
- S_p: use the effective field size *incident on the patient* projected to the point or normalization depth.[1]
- TPR and TMR: same field-size logic as S_p, at the calculation depth.[1]
- PDD: use the field size incident on the patient, typically defined at the surface for SSD formalism.[1]
- WF: for wedged irregular fields, use the equivalent square of the *irregular exposed field*.[1]

4.2 Varian versus Elekta for S_c

Machine-Design Comparison

- Varian tertiary MLC: use jaw/collimator field size for S_c
- Elekta upper-jaw-replacement MLC: MLC-defined aperture can determine S_c
- Reason: S_c depends on the aperture system closest to the source/head-scatter source

Figure 4: Machine-design dependence of the S_c field-size argument.

Oral-exam answer

What field size do you use for S_c on Varian versus Elekta, and why?

For a **Varian tertiary MLC** design, TG-71 says the tertiary MLC behaves like a block for S_c, so the S_c field-size argument is the *collimator jaw field size*. The reason is that the jaw system remains the aperture closest to the target that dominates head-scatter behavior.[1]

For **Elekta/Philips upper-jaw-replacement** MLC designs, TG-71 says S_c can be represented by the equivalent square of the *MLC-blocked field area*, because the MLC is itself replacing an upper jaw and therefore changes the visible source and head scatter more directly.[1]

4.3 Equivalent Square and Equivalent Circle

For rectangular fields, an equivalent square field size is often estimated using the Sterling approximation:

$$S_{eq} = 4\frac{A}{P} \tag{24}$$

where A is the field area and P is the field perimeter. This approximation works because scatter depends on both the irradiated area and the amount of field edge.

For a rectangular field:

$$A = LW$$

$$P = 2L + 2W = 2(L + W)$$

Substituting:

$$\begin{aligned} S_{eq} &= 4\frac{A}{P} && (25)\\ &= 4\frac{LW}{2(L+W)} && (26)\\ &= \frac{2LW}{L+W} && (27) \end{aligned}$$

Therefore:

$$\boxed{S_{eq} = \frac{2LW}{L+W}} \tag{28}$$

For an equivalent circular field with the same area as the rectangular field:

$$A_{\text{circle}} = A_{\text{rectangle}}$$

$$\pi r_{eq}^2 = LW$$

Solving for r_{eq}:

$$\boxed{r_{eq} = \sqrt{\frac{LW}{\pi}}} \tag{29}$$

and the equivalent circular diameter is:

$$\boxed{d_{eq} = 2\sqrt{\frac{LW}{\pi}}} \tag{30}$$

Important: S_{eq} is the side length of an equivalent square, while r_{eq} is the radius of an equal-area circle. with jaw settings defined at isocenter.[1]

Common trap

Collimator-exchange effect

The $4A/P$ approximation does *not* model collimator exchange. A 5×40 field and a 40×5 field have the same $4A/P$, but not necessarily the same head scatter. TG-71 notes that the resulting error is often small ($< 2\%$) but should be evaluated for your machine, especially for large aspect ratios.[1]

4.4 How S_c, S_p, and $S_{c,p}$ change with field size

Oral-exam answer

How do S_c, S_p, and $S_{c,p}$ change with field size?

- S_c increases with larger collimator opening because more of the extra-focal/head-scatter source is visible.[1]
- S_p increases with larger irradiated field because more phantom scatter contributes to the point.[1]
- $S_{c,p} = S_c \times S_p$, so total output generally increases with field size.[1]

Qualitatively: small fields $< 10 \times 10$ have factors below 1.0, 10×10 is normalized to 1.0, and larger fields rise above 1.0.

High-yield point

What changes faster with field size?

For many beams, S_c shows a stronger head-design dependence, while S_p is more beam-quality and depth dependent and more "patient-side" in interpretation. A great oral-exam sentence is:

S_c is machine/head scatter; S_p is phantom scatter; $S_{c,p}$ is what I measure in water if I do not separate them.

4.5 How PDD changes with field size and SSD

Oral-exam answer

How does PDD change with field size and SSD?

- **With larger field size**: PDD usually increases at depth because additional scattered photons contribute dose further downstream.[6, 1]
- **With larger SSD**: PDD usually increases because the geometric falloff from $d_{\max}$ to depth d is reduced; Mayneord's factor captures the inverse-square part of that effect.[5, 6]

4.6 How to handle blocked fields

Oral-exam answer

Where do you put the dose point, and what field size do you use for blocked fields?

Put the dose point in an *open, high-dose, low-gradient, soft-tissue region* if at all possible. TG-114 says to avoid points near field edges or interfaces, and points within roughly 2 cm of a field edge should be treated cautiously because of disequilibrium effects.[2]

For field sizes:

- use S_c field-size logic based on machine design,
- use the *effective field incident on the patient* for S_p, TPR, TMR, and PDD,
- use the irregular exposed-field equivalent square for WF in wedged blocked fields.[1]

If the central axis is blocked or the comparison point must be off-axis, TG-71 recommends measurement support if the geometry is complex or outside your tabulated data.[1]

5 Heterogeneity corrections and when MU checks fail

5.1 Does the second check include heterogeneity?

Oral-exam answer

Does MU check include heterogeneity calculations?

It *should* if the primary calculation includes heterogeneity. TG-114 strongly recommends that when the primary calculation uses heterogeneity correction, the verification calculation should also use heterogeneity correction, even if the secondary method is simpler and uses a different correction model.[2]

Common trap

A dangerous but common weak practice

Older point-dose second checks often ignored heterogeneity entirely. That may be acceptable only as a very rough plausibility screen, but in thorax, near lung interfaces, or in small fields it can create disagreements that are so large that the comparison stops being clinically meaningful.[2, 4]

5.2 Physical depth versus radiological depth

Definition

Radiological depth

A first-order heterogeneity correction starts by replacing the physical path length with an electron-density weighted path length:

$$d_{\text{rad}} = \int_{\text{ray line}} \rho_e(s)\, ds \approx \sum_i \rho_{e,i}\, \Delta s_i. \tag{31}$$

For point-dose hand checks, this is often the single most important heterogeneity quantity.[2, 4]

5.3 Simple photon heterogeneity formulas

Key formula

RTAR-style correction used in simple hand checks

For a heterogeneity-corrected photon hand check, TG-114 gives the simple ratio-of-TPR form

$$CF_{\mathrm{RTAR}} = \frac{TPR(d_{\mathrm{eff}}, r_d)}{TPR(d, r_d)}. \tag{32}$$

Key formula

General definition of heterogeneity correction factor

$$CF = \frac{\text{dose in heterogeneous medium}}{\text{dose at same point in homogeneous medium}}. \tag{33}$$

Key formula

Batho / power-law style idea

For slab-like heterogeneities, Batho-type corrections modify the primary/path-length term and approximately correct scatter by a power law in density. In oral exams, it is usually enough to say:

Batho is still a useful hand-calculation approximation for slab heterogeneity, but it is limited because it does not model 3D lateral disequilibrium well.

5.4 Rule-of-thumb thinking for lung and bone

Oral-exam answer

How do you do heterogeneity calculations, and what is the rule-of-thumb for lung versus tissue?

Step 1: Estimate radiological depth along the beam path.

Step 2: For a *large, fairly uniform field* and a point not too close to an interface, use a radiological-depth-based correction such as Eq. (32) as a first-order estimate.[2]

Rule of thumb for lung: if the point lies *beyond* a broad lung slab, replacing tissue by lung lowers radiological depth, so a simple path-length model usually predicts *more* transmitted dose and therefore *fewer* MUs than a homogeneous calculation. But that intuition can fail in small fields or within the lung because lateral electron equilibrium is lost. TG-65 explicitly notes that for a 5×5 field in lung, the correction factor can fall below 1 due to disequilibrium.[4]

Rule of thumb for bone / dense media: radiological depth is larger, so distal dose usually drops and you need *more* MU for the same target-point dose. But near interfaces there can be build-down and rebuild-up, so a simple slab correction can be misleading.[4]

High-yield point

Best oral-exam phrasing

Large-field points well inside tissue are path-length problems. Small fields and interface points are not just path-length problems; they are charged-particle-equilibrium problems.

5.5 How big can heterogeneity effects be?

TG-114 notes that in common clinical sites such as breast or prostate, heterogeneity corrections are often small, but in thoracic irradiation they can be substantial.[2] TG-65 reports that for simple mediastinal/lung geometries, target-point corrections can range roughly from 0.95 to 1.16, and can be even larger in oblique or lateral fields.[4]

TG-114 also notes that simple photon scatter corrections to account for patient geometry and reduced scatter can commonly change the MU by about 2% to 8%, with even larger effects for very small fields.[2]

5.6 When do you expect a second MU check to fail or give a bad result?

Oral-exam answer

A strong exam answer is to separate *physics-limited* failures from *process-limited* failures.

Physics-limited / algorithm-limited situations

- point near field edge or in penumbra,
- point near tissue interface,
- small fields in lung,
- high-gradient off-axis wedged points,
- blocked central axis,
- highly irregular electron cutouts,
- extended-SSD electrons with very small inserts,
- highly modulated IMRT/VMAT if the secondary check is only point-dose based.[1, 2, 3, 4]

Process-limited situations

- wrong SSD or depth imported from the TPS,
- wrong CT dataset,
- wrong density table or contrast artifact,
- omitted bolus or wrong tray/wedge,
- wrong wedge orientation,
- wrong field-size interpretation for S_c vs S_p,
- correlated beam-model errors between TPS and second-check system.[2, 3]

5.7 Tolerance for TPS versus second MU check

5.7.1 Non-IMRT photon and electron plans

TG-114 emphasizes that each clinic must establish its own documented action levels during commissioning, but provides widely used guidance.[2] A compact version is shown below.

Table 2: Compact TG-114-style action-level summary for non-IMRT comparisons.[2]

Situation	**Similar algorithms**	**Different algorithms**
Minimal field shaping, homogeneous	$\sim$ 2% to 3%	$\sim$ 2.5% to 3%
Substantial shaping / contour change	$\sim$ 2.5% to 4%	$\sim$ 3% to 4%
Wedged or off-axis, homogeneous	$\sim$ 2% to 3%	$\sim$ 3.5% to 5%
Large field with heterogeneity correction	$\sim$ 2% to 3%	$\sim$ 2.5% to 3.5%
Wedged/off-axis with heterogeneity correction	$\sim$ 2% to 3%	$\sim$ 3.5% to 4.5%
Small field and/or low-density heterogeneity	$\sim$ 3% to 3.5%	$\sim$ 4% to 5%

5.7.2 IMRT and VMAT

Oral-exam answer

What tolerance do you quote for IMRT/VMAT?

TG-219 says every IMRT/VMAT plan should receive an independent secondary dose/MU calculation, at least in 1D and preferably in 2D/3D, and that evaluation should be based on the *composite plan* rather than isolated single beams.[3]

For 1D point-dose style checks, a commonly quoted TG-219 benchmark is ±5% in a high-dose, low-gradient homogeneous region for the *composite plan*. For 2D/3D comparisons, TG-219 points users to TG-218-type gamma criteria; a commonly cited action level is about 90% passing at 3%/2 mm, again assessed on the composite plan.[3]

5.8 How to handle a failing MU check

Clinical workflow

If the MU check fails, do this in order

1. **Stop and verify the comparison point.** Is it near an interface, field edge, mandibular air cavity, tiny lung segment, or steep gradient?
2. **Verify the prescription and beam setup.** Dose, energy, SSD/SAD, beam weight, wedge angle/orientation, tray, bolus, cutout, skin collimation, gantry/collimator/couch assumptions.
3. **Verify field-size logic.** Especially S_c versus S_p arguments, blocked-field equivalent square, and Varian-versus-Elekta treatment of S_c.
4. **Check heterogeneity handling.** Was it included in both systems? Was the radiological path reasonable?
5. **Look for correlated-data problems.** Is the second check using the same beam data, same density table, or same imported bad point?
6. **Measure if needed.** If the point is complex, move to a measurement or a more robust 3D independent calculation.
7. **Escalate if systematic.** Repeated patterns suggest beam-model, data-transfer, or software issues and may require vendor support or recommissioning.[2, 3]

6 Measuring and commissioning the ingredients

6.1 Photon measurements

6.2 Electron measurements

6.3 How to measure $S_{c,p}$

Oral-exam answer

How do you measure $S_{c,p}$?

Measure it *in a water phantom* at SSD_0 with the detector at the normalization depth d_0 for a series of field sizes. Then separate phantom scatter by dividing by the independently measured in-air S_c:

$$S_p(r) = \frac{S_{c,p}(r)}{S_c(r)}. \tag{34}$$

If your clinic's d_0 is 10 cm, your $S_{c,p}$ and S_p tables should be consistent with that depth, because S_p is depth dependent.[1]

Table 3: Practical measurement summary for photon MU ingredients.[1]

Quantity	Where / how measured	Exam-ready note
$\dot{D}_0$	Water phantom, normalization conditions	Local reference output for the beam.
S_c	Mini-phantom in air, detector at isocenter, TG-71 recommends a 4 cm diameter mini-phantom with detector at 10 cm depth	Remove nearby scatter; beam toward a wall rather than floor if possible; watch stem/cable effects, especially for long rectangles.
$S_{c,p}$	Water phantom at SSD_0, detector at d_0, for different collimator settings	This is the total scatter factor in water.
S_p	Usually computed as $S_{c,p}/S_c$	Use the irradiated field at the point, not blindly the jaw field.
WF	Water phantom, chamber at isocenter, measure versus field size and depth	TG-71 recommends chamber axis perpendicular to wedge gradient and measurements with wedge in opposite orientations.
TF	Water phantom, beyond contamination-electron region	Do not measure tray factor in buildup and then use it at depth.
OAR	Prefer primary off-axis ratio data	Raw profile data near field edges can be misleading if you use it as OAR.
PDD	Water phantom, fixed SSD	Depends on energy, field size, SSD.
TPR/TMR	Water phantom, SAD/isocentric geometry	Field size defined at calculation depth.

Table 4: Practical measurement summary for electron MU ingredients.[1]

Quantity	Where / how measured	Exam-ready note
$\dot{D}_0$	Electron reference geometry in water	Output anchor for the beam.
S_e	Depth d_m, for each applicator/insert at nominal SSD	Output is primarily applicator + insert dependent.
PDD	Water phantom depth-dose for the insert / field	For electrons, PDD is usually determined mainly by insert size; with skin collimation, use *skin-collimation field size* for the PDD term.[1]
SSD_{eff}	Fit output versus gap over clinical SSD range	Energy and field-size dependent.
f_{air}	Measured output at extended SSD relative to inverse-square estimate	Captures side-scatter loss; more severe for smaller inserts and lower energies.

6.4 How to measure WF

Oral-exam answer

How do you measure wedge factor?

For a **physical wedge**, TG-71 recommends measuring WF as a function of *field size and depth*. Put the chamber at isocenter, orient the chamber axis perpendicular to the wedge-gradient direction, and repeat with the wedge in both orientations to reduce setup bias.[1]

For a **dynamic / nonphysical wedge**, understand the vendor-specific behavior. Field-size and off-axis dependence can differ from a physical wedge, and enhanced dynamic wedge factors depend strongly on the moving-jaw/fixed-jaw geometry.[1]

6.5 Typical values worth remembering

Table 5: Good numbers to remember for oral exams. These are examples, not universal constants.[1, 2, 4]

Factor	What to say	Useful example
$S_{c,p}$	By definition, 10×10 is 1.0; smaller fields are usually below 1.0 and larger fields above 1.0.	Say trend, not a machine-independent number.
WF	Strongly angle dependent; also field-size and depth dependent for physical wedges.	TG-71 sample 6 MV, 10×10 near $d_{\max}$: about 0.707 (15°), 0.539 (30°), 0.480 (45°), 0.396 (60°).
TF	Usually close to 1.0 but *must be measured locally.*	A few percent attenuation is common clinically, but use your machine's data.
CF	Often small in homogeneous sites, potentially large in chest.	TG-65 reports thoracic target-point corrections roughly 0.95 to 1.16 in simple geometries.
f_{air}	Equals 1.0 at nominal SSD and drops with gap, more for small fields.	TG-71 sample 9 MeV, 2×2 field: 0.882 at 105 cm, 0.505 at 120 cm.

6.6 Electron cutouts between tabulated data

Oral-exam answer

How do you handle electron cutouts whose dimensions lie between tabulated data?

For **square** inserts, interpolate between measured data points in cutout size.

For **rectangular** inserts, TG-71 recommends the *square-root (geometric-mean) method* rather than the simple equivalent-square method:

$$S_e(r_a, L \times W) = \sqrt{S_e(r_a, L \times L)\, S_e(r_a, W \times W)}, \tag{35}$$

and the same approach is recommended for PDD with a final renormalization if d_m shifts.[1]

For **irregular** cutouts, the best answer is:

- measure directly if the field is unusual or used often;
- otherwise approximate by a rectangle built around the broadest region that governs output, using clinical judgment and previous data;
- be especially cautious for small inserts, high energy, and extended SSD because interpolation errors can become clinically important.[1]

6.7 Skin collimation

Oral-exam answer

What happens if skin collimation is used for electrons?

TG-71 says the output is mainly governed by applicator and insert size, but the *PDD* is mainly determined by the *skin-collimation field size*. Therefore, when skin collimation is present, the field-size argument for the PDD term should be the skin-collimation field size, not just the insert/applicator size.[1]

6.8 Extended SSD equations for photons and electrons

Oral-exam answer

Photon extended SSD summary

Use the SSD/PDD formalism with the explicit inverse-square term

$$\left(\frac{SSD_0 + d_0}{SSD + d_0}\right)^2$$

and recognize that PDD itself changes with SSD. Mayneord's factor is the classic geometry-only conversion when measured PDD data at the clinical SSD are unavailable.[1, 5]

Oral-exam answer

Electron extended SSD summary

Use either the effective SSD method or the air-gap-factor method. In either case, *do not* assume electrons follow pure inverse square; they lose side scatter and therefore often fall off faster than $1/r^2$, particularly for small fields and low energies.[1]

7 TPS versus second-check software, including RadCalc and Mobius

7.1 What TG-114 and TG-219 expect from independence

Oral-exam answer

What makes a second check truly independent?

TG-114 says independence means more than "a second number." The calculation program should be separate from the primary TPS, and the beam data / parameter files should also be separate and independent. Even if patient parameters are electronically imported, the physicist should independently verify that the imported geometry and point selection actually make sense.[2]

TG-219 makes the same point for IMRT/VMAT: independence can come from independent algorithms and/or independent beam data, and the preferred situation is that both are independent.[3]

7.2 How TPS problems affect the second check

Common trap

Correlated-error problem

If the TPS and the secondary system use the same beam data, same density mapping assumptions, strongly similar algorithms, and the same imported bad comparison point, then the second check can agree beautifully with the TPS and still be wrong.[2, 3]

Oral-exam answer

If there is an issue with the TPS, how would that affect the second check?

It depends on the kind of issue.

- If the TPS problem is a missing wedge, wrong SSD, wrong point, wrong bolus, or wrong density override, an independent second check may catch it *if those parameters are independently verified.*
- If the TPS problem is a beam-model or data-table issue, a second system with independent beam data has a better chance of catching it.
- If the secondary system shares the same beam data or closely shares the same algorithmic assumptions, then the error can be correlated and therefore missed.[2, 3]

7.3 RadCalc

Oral-exam answer

What should you know about RadCalc?

Historically, RadCalc has been widely used as an independent MU / point-dose check system. The official RadCalc technique documentation states that for IMRT it performs independent MU or point-dose calculations using a *modified Clarkson scatter integration* together with a head-scatter algorithm.[9, 10]

Modern RadCalc also offers CT-based **3D** verification modules based on **collapsed cone convolution-superposition** and **Monte Carlo**, allowing dose-volume verification on the patient's planning CT rather than only a single-point plausibility check.[11]

Common trap

Oral-exam nuance

If you are talking about *legacy point-dose RadCalc*, say that it is strongest as a plausibility check but can struggle in severe heterogeneity, small fields, high gradients, and heavy modulation.

If you are talking about *RadCalc 3D CCCS/MC*, say that it is much closer to a modern independent volumetric dose verification workflow.[10, 11]

7.4 Mobius and Fontenot's paper

Oral-exam answer

What should you know about Mobius / Mobius3D?

Mobius3D is an independent 3D dose verification system that reads the treatment plan and recalculates dose on the patient CT using its own collapsed-cone convolution-superposition algorithm.[12, 13]

Fontenot's 2014 paper is important because it framed Mobius3D as a more modern secondary-check tool: instead of only a single point, it performs an independent 3D dose verification and can also automatically compare DVHs against reference benchmarks.[8]

Oral-exam answer

What did Fontenot find?

Fontenot evaluated Mobius3D against phantom measurements and TPS calculations for IMRT/VMAT plans. The paper reported that Mobius3D agreed with measurement comparably to the TPS in phantom tests, and for patient VMAT plans the average agreement with the TPS was high, with about 99.5% of dose points satisfying $\gamma_{5\%,3\mathrm{mm}} < 1$. The system also identified several plans that exceeded protocol-based DVH benchmarks.[8]

Common trap

What not to say

Do *not* say Mobius or any secondary-check system makes measurement-based QA unnecessary. TG-219 is explicit that independent secondary calculations should complement, not replace, the larger IMRT/VMAT QA program.[3]

7.5 TPS versus second MU check

Oral-exam answer

How do you compare TPS versus second MU check in one sentence?

The TPS is the primary clinical dose engine; the second MU check is an independently commissioned plausibility test of that answer, ideally with independent beam data and a different algorithm, and interpreted in the context of point selection, heterogeneity, and the known limits of both systems.

8 Rapid-fire oral exam answers

8.1 The shortest answers you should be able to give cleanly

Oral-exam answer

Know parameters and equations, definitions of parameters, including BSF

I know the photon MU equations in both TPR and PDD form, the electron MU equations at nominal and extended SSD, and the meanings of S_c, S_p, $S_{c,p}$, WF, TF, OAR, r_c, r_d, r_{d_0}, SSD_{eff}, and f_{air}. BSF is a classical backscatter factor / peak scatter concept used more naturally in older TAR-based or kilovoltage dosimetry, whereas modern MV hand calculations usually use S_c, S_p, and $S_{c,p}$ instead.[1, 6]

Oral-exam answer

TG-114

TG-114 is the AAPM report for independent MU verification in non-IMRT clinical radiotherapy. Its key messages are that second checks still matter, they are only one part of full plan review, independence requires a separate system and separate beam data whenever possible, point placement matters enormously, and action levels should be established locally from commissioning rather than copied blindly.[2]

Oral-exam answer

TG-219

TG-219 extends the idea to IMRT/VMAT. It recommends a secondary dose/MU calculation for every IMRT/VMAT plan, at least in 1D and preferably in 2D/3D, with acceptability based on the *composite plan*. It also emphasizes that this does not replace measurement-based PSQA.[3]

Oral-exam answer

Mayneord factor

Mayneord's factor converts PDD from one SSD to another by accounting only for the inverse-square component of the change. It is derived by taking the ratio of PDD definitions at two SSDs and assuming the scatter/attenuation ratio is unchanged. It is useful for hand checks, but can be wrong by several percent at very extended SSD because scatter conditions also change.[5, 6]

Oral-exam answer

PDD to TMR conversion

PDD and TMR differ because PDD has the geometric $1/r^2$ effect built in, while TMR compares doses at equal source-to-detector distance in phantom. To convert PDD to TMR, remove the geometry using $\left((SSD+d)/(SSD+d_{\max})\right)^2$ and correct for the change in phantom scatter between the field at $d_{\max}$ and the field at d.[1, 7]

Oral-exam answer

Inverse square law, extended SSD versus 100 cm

For photons, extended SSD is handled explicitly in the MU equation by the source-to-reference-point distance ratio squared, and the PDD itself changes with SSD. For electrons, extended SSD is not pure inverse square because side scatter is lost, so you need SSD_{eff} or f_{air}.[1]

Oral-exam answer

TPS versus second MU check

The TPS is the primary planning engine; the second check is an independently commissioned plausibility check. Strong agreement is reassuring, but disagreement must be interpreted in the context of point placement, heterogeneity, field modulation, and correlated beam-data assumptions.[2, 3]

Oral-exam answer

When do you expect the MU check to fail or give a bad result?

Near interfaces, near field edges, in small fields, in low-density media, in off-axis wedged geometries, in blocked central-axis fields, in highly irregular electron cutouts, in extended-SSD electrons with tiny inserts, and in IMRT/VMAT if the secondary method is only a simplistic single-point algorithm.[1, 2, 4, 3]

Oral-exam answer

Tolerance for TPS versus second MU check

For non-IMRT, TG-114 suggests action levels that widen with complexity; roughly 2% is possible in simple homogeneous geometry and 4% to 5% may be appropriate in wedged/off-axis or heterogeneity-corrected small-field situations depending on method similarity.[2]

For IMRT/VMAT, TG-219 commonly gets summarized as about ±5% for a 1D composite-plan point check in a high-dose low-gradient homogeneous region, and approximately 90% passing at 3%/2 mm for a 2D/3D style comparison, though each clinic should validate its own action levels.[3]

Oral-exam answer

What if I do not have an MU check?

Then I need another valid independent verification method, usually an independent manual/system calculation or an in-phantom measurement. TG-114 also says the department should have a written emergency policy that limits how much dose can be delivered before full physics review.[2]

Oral-exam answer

What if there is an issue with the TPS? How would it affect the second check?

If the second check is truly independent in algorithm and beam data, it may catch the problem. If it shares the same beam model, same density assumptions, or the same bad imported geometry, the two systems can agree on the same wrong answer.[2, 3]

Oral-exam answer

Does MU check include heterogeneity calculations?

It should if the primary plan uses them. TG-114 strongly recommends that heterogeneity-corrected primary calculations be compared with heterogeneity-corrected verification calculations, even if the methods differ.[2]

Oral-exam answer

How much do you expect your MU calculation to change if factors change?

For multiplicative factors it is roughly one-for-one in the opposite direction. A 1% change in S_c, S_p, PDD, TMR, WF, or TF changes MU by about 1% in the opposite direction. Distance errors can be about $2\Delta R/R$ from inverse square alone, so they can be very important.[1]

Oral-exam answer

How would you handle a failing MU calc?

First verify the point, then verify geometry and accessories, then verify field-size logic, then verify heterogeneity handling, then consider measurement or a more robust 3D recalculation. Systematic patterns mean I should suspect data transfer, beam modeling, or software issues and escalate appropriately.[2, 3]

Oral-exam answer

What field size do you use for S_c on Varian versus Elekta and why?

Varian tertiary MLC: use jaw/collimator field for S_c because the tertiary MLC behaves like a block for head-scatter purposes. Elekta upper-jaw-replacement MLC: use the equivalent square of the MLC-blocked field because that aperture is the relevant source-proximal scatter-defining system.[1]

Oral-exam answer

How do S_c, S_p, and $S_{c,p}$ change with field size?

All generally increase with increasing field size, but for different reasons: S_c because more head scatter is visible; S_p because phantom scatter volume is larger; $S_{c,p}$ because it is both together.[1]

Oral-exam answer

How do you handle blocked fields?

Use a comparison point in open, high-dose, low-gradient tissue if possible. Use machine-design logic for S_c, but use the effective field incident on the patient for S_p, depth-dose factors, and wedge-factor arguments. If the geometry is off-axis or the CAX is blocked, support the calculation with measurement if necessary.[1, 2]

Oral-exam answer

How do you handle electron cutouts between tabulated data?

Interpolate for squares, use the square-root/geometric-mean method for rectangles, and measure directly or use prior validated data for irregular cutouts if they are clinically important or outside the well-behaved interpolation range.[1]

Oral-exam answer

What are typical values for $S_{c,p}$, WF, CF, etc.?

$S_{c,p}$ is normalized to 1.0 at 10×10 and generally runs lower for smaller fields and higher for larger ones. WF can be remembered roughly as 0.71, 0.54, 0.48, 0.40 for 15°, 30°, 45°, 60° in a classic 6 MV 10×10 example. CF is often near 1.0 in homogeneous sites but can depart substantially in chest/lung geometry, with simple thoracic target-point corrections roughly 0.95 to 1.16.[1, 4]

9 Final high-yield checklist

High-yield point

If you can say the following things clearly, you are in excellent shape for the oral exam:

1. I know the photon MU equation in both TPR and PDD form.
2. I know the electron MU equation at nominal and extended SSD.
3. I know exactly which field size goes into S_c, S_p, and depth-dose terms.
4. I can derive Mayneord's factor from the PDD definition.
5. I can derive PDD to TMR conversion and explain the scatter correction.
6. I know why electrons at extended SSD are not pure inverse square.
7. I know when a second MU check is meaningful and when it is weak.
8. I know TG-114 action-level philosophy and TG-219 composite-plan philosophy.
9. I know the practical differences between point-dose tools and modern 3D secondary systems such as RadCalc 3D and Mobius3D.
10. I know how to triage a failing MU check without hand-waving.

Common trap

The single best oral-exam closing line

A second MU check is only as good as its independence, its commissioning, and its comparison point.

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